

Update to the hake Reference Case Operating Model with corrected longline data, and 2021 commercial and 2022 survey data

A. Ross-Gillespie¹

Contact email: andrea.ross-gillespie@uct.ac.za

Summary

The hake Reference Case Operating Model is updated with two iterations: first the post-2015 longline catch data have been updated following some data validation and second the model is further updated to include the 2021 commercial catch and CPUE data as well as the 2022 survey abundance data. The impact of both changes on the assessment results is minimal. The updated Reference Case results suggest that the *M. paradoxus* stock is on a downward swing (the B^{sp}/B_{MSY} estimate is 1.60 for 2021 and 1.50 for 2022), while *M. capensis* appears to continue on an upward trajectory (the B^{sp}/B_{MSY} estimate is 2.63 for 2021 and 2.81 for 2022).

Keywords: South African hake, Reference Case update, 2022 data

Introduction

Results are provided for three iterations of the Reference Case (RC) Operating Model (OM).

Run1: The RC as used for the development of OMP-2022, with commercial data up to 2020 and survey data up to 2021 (Ross-Gillespie and Butterworth, 2022).

Run2: The same as (1), but with the post-2015 longline catches updated following some data validation.

Run3: An update of (2), with the 2021 commercial catch and CPUE data and the 2022 survey abundance estimates included in the model fit.

The results for Run2 have been included to determine the impact of the longline catch data update on the assessment results. Run2 and Run3 should then be compared to ascertain the impact of the 2022 data update.

Table 1 lists the original longline catches (used for Run1) alongside the updated catches (used for Run2 and Run3). Table 2 lists the updated commercial catches for all the fleets from 1978 to 2021 used for Run3. Table 3 lists the updated GLM CPUE data used for Run3 and Table 4 lists the survey abundance estimates including the most recent 2022 West Coast summer survey estimates.

Results

Table 5 lists key assessment outputs for the three iterations of the RC OM. Table 6a and b list the negative log-likelihood components for the three models, in (a) absolute terms and (b) relative to Run1.

Figure 1 - Figure 6 show various assessment outputs for the three model fits.

¹ Marine Resource Assessment and Management Group, University of Cape Town, Rondebosch, 7701

Discussion

Updating the longline catch data

Updating the longline catch data decreases the B/B_{MSY} slightly for both species, but the difference is minimal, and the impact on the negative log-likelihood is negligible.

Inclusion of the 2021 commercial catch and CPUE data, and 2022 survey abundance estimates

The results for the RC run with the updated data are consistent with those from before the update. The B^{sp}/B_{MSY} estimate for 2021 is slightly lower for *M. paradoxus* (1.63 for Run2 and 1.60 for Run3, see Table 5) and slightly higher for *M. capensis* (2.61 for Run2 and 2.63 for Run3), but this difference is minimal. The 2022 estimates suggest that the *M. paradoxus* stock is on a downward swing reasonably consistent with oscillations evident for preceding years, while *M. capensis* appears to continue on an upward trajectory.

References

- Ross-Gillespie, A. and Butterworth, D.S. 2022. Corrected Reference Set conditioning results and updated CMP projection results. DEFF Fisheries document: FISHERIES/2022/OCT/SWG-DEM/26: 25pp.
- Fairweather, T.P. 2022. Standardised abundance estimates for key commercial species 2016-2022. DEFF Fisheries document: FISHERIES/2022/SEP/SWG-DEM/19: 2pp.

Table 1: List of the original longline catches as uses for Run1, next to the updated catches as used for Run2 and Run3, as provided by D. Durholtz (*pers. comm.*).

	Original longline catches				Updated longline catches			
	<i>M. paradoxus</i>		<i>M. capensis</i>		<i>M. paradoxus</i>		<i>M. capensis</i>	
	WC	SC	WC	SC	WC	SC	WC	SC
2015	4.001	0.018	2.325	0.053	5.205	0.023	3.025	0.069
2016	2.806	0.001	4.360	0.002	3.697	0.001	5.745	0.003
2017	5.288	0.025	2.807	0.126	5.300	0.025	2.813	0.126
2018	5.156	0.089	2.615	0.481	5.217	0.090	2.646	0.487
2019	3.177	0.020	2.160	0.179	5.328	0.034	3.623	0.299
2020	3.220	0.003	1.293	0.177	5.847	0.047	2.348	0.321

Table 2: Species-disaggregated catches (in thousand tons) by fleet of South African hake from the south and west coasts for the period 1978-present. The 2021 catches were provided by D. Durholtz (*pers. comm.*).

	<i>M. paradoxus</i>				<i>M. capensis</i>					
	Offshore		Longline		Offshore		Inshore	Longline		Handline
	WC	SC	WC	SC	WC	SC	SC	WC	SC	SC
1978	101.042	3.220	-	-	26.470	4.365	4.931	-	-	-
1979	94.331	1.924	-	-	39.192	4.995	6.093	-	-	-
1980	99.654	2.206	-	-	33.873	4.254	9.121	-	-	-
1981	88.883	0.910	-	-	32.048	4.575	9.400	-	-	-
1982	83.618	3.353	-	-	29.732	8.005	8.089	-	-	-
1983	71.238	4.723	0.126	-	23.195	7.792	7.672	0.104	-	-
1984	82.358	3.796	0.200	0.005	28.897	7.139	9.035	0.166	0.011	-
1985	94.428	8.059	0.638	0.091	30.642	11.957	9.203	0.529	0.201	0.065
1986	103.756	8.580	0.753	0.094	30.049	7.385	8.724	0.625	0.208	0.084
1987	93.517	7.460	1.952	0.110	24.008	8.225	8.607	1.619	0.243	0.096
1988	79.913	5.876	2.833	0.103	26.669	8.640	8.417	2.350	0.228	0.071
1989	82.230	6.182	0.158	0.010	25.029	12.730	10.038	0.132	0.022	0.137
1990	81.996	9.341	0.211	-	21.640	13.451	10.012	0.175	-	0.348
1991	87.093	12.448	-	0.932	19.357	9.626	8.206	-	2.068	1.270
1992	84.768	17.297	-	0.466	18.519	9.165	9.252	-	1.034	1.099
1993	102.125	9.880	-	-	15.940	4.380	8.870	-	-	0.278
1994	103.541	6.726	0.882	0.194	20.327	4.326	9.569	0.732	0.432	0.449
1995	100.268	4.004	0.523	0.202	20.629	3.146	10.630	0.434	0.448	0.756
1996	107.381	8.966	1.308	0.568	21.794	4.323	11.062	1.086	1.260	1.515
1997	100.654	10.509	1.410	0.582	16.500	5.327	8.834	1.170	1.290	1.404
1998	111.154	9.742	0.505	0.457	16.499	4.411	8.283	0.419	1.014	1.738
1999	88.581	11.420	1.532	1.288	15.179	3.926	8.595	1.272	2.856	2.749
2000	96.587	7.700	2.706	3.105	21.114	5.830	10.906	2.000	1.977	5.500
2001	101.247	7.850	1.417	0.084	16.349	8.306	11.836	2.394	1.527	7.300
2002	91.207	12.443	4.469	1.585	13.724	6.141	9.581	2.391	2.546	3.500
2003	93.711	17.397	3.305	1.252	11.665	7.636	9.883	2.526	3.078	3.000
2004	85.722	26.065	2.855	1.196	12.510	8.704	10.004	2.297	2.731	1.600
2005	85.869	21.778	3.091	0.472	9.398	7.468	7.881	2.773	3.270	0.700
2006	81.513	18.050	3.241	0.485	11.984	6.578	5.524	2.520	3.227	0.400
2007	92.724	13.488	2.512	3.021	16.145	3.757	6.350	2.522	2.522	0.400
2008	85.538	13.191	2.255	0.809	13.838	4.316	5.496	1.937	1.893	0.231
2009	68.202	10.895	2.410	1.069	12.296	4.806	5.639	2.828	2.520	0.265
2010	69.709	15.457	2.394	1.527	10.186	4.055	5.472	3.086	3.024	0.275
2011	76.576	17.904	2.522	0.140	15.673	4.086	6.013	3.521	3.047	0.186
2012	81.411	16.542	4.358	0.306	12.928	4.584	3.223	2.570	1.737	0.008
2013	74.341	28.859	6.056	0.060	8.761	4.475	2.920	2.606	1.308	0.000
2014	73.252	41.156	6.879	0.008	9.671	6.286	2.965	2.123	0.315	0.001
2015	77.521	31.745	5.205	0.023	12.727	4.035	3.077	3.025	0.069	0.001
2016	93.173	18.968	3.697	0.001	14.744	2.810	3.973	5.745	0.003	0.001
2017	72.326	30.961	5.300	0.025	15.273	4.466	2.812	2.813	0.126	0.004
2018	64.252	29.218	5.217	0.090	12.689	12.863	3.983	2.646	0.487	0.024
2019	70.608	22.201	5.328	0.034	14.193	9.454	4.149	3.623	0.299	0.009
2020	97.093	10.061	5.847	0.047	18.115	3.500	4.536	2.348	0.321	0.004
2021	102.865	15.597	5.892	0.018	15.585	2.937	4.517	2.932	0.194	0.010

Table 3: GLM standardized CPUE data for *M. paradoxus* and *M. capensis*, corresponding to the Model A6b species splitting algorithm provided by J. Glazer (*pers. comm.*).

Year	GLM CPUE (kg min ⁻¹)			
	<i>M. paradoxus</i>		<i>M. capensis</i>	
	West Coast	South Coast	West Coast	South Coast
1978	9.16	1.73	2.34	4.04
1979	9.12	1.45	3.24	4.49
1980	8.62	2.52	2.91	5.46
1981	8.51	1.70	3.03	4.71
1982	8.23	2.38	2.62	4.96
1983	9.17	2.40	3.27	5.84
1984	9.35	2.66	3.32	6.78
1985	11.00	3.49	3.73	8.54
1986	9.50	3.52	3.09	7.02
1987	7.98	3.23	2.60	6.45
1988	7.72	2.63	2.38	6.57
1989	8.40	2.73	2.55	7.33
1990	8.88	3.52	2.28	8.83
1991	9.85	3.90	2.77	7.89
1992	8.75	4.30	3.16	7.41
1993	8.88	3.70	3.06	5.48
1994	9.57	3.30	3.59	6.55
1995	8.45	2.55	3.45	6.04
1996	9.40	3.63	3.97	6.04
1997	8.28	3.92	3.36	5.04
1998	9.39	3.71	3.83	4.98
1999	7.71	4.29	3.41	5.38
2000	6.98	3.44	3.21	6.00
2001	5.59	3.39	2.27	4.52
2002	5.45	3.04	2.16	4.85
2003	6.31	3.52	1.94	5.19
2004	5.16	2.96	1.80	4.31
2005	5.01	2.53	1.37	3.72
2006	5.33	2.66	1.57	2.93
2007	6.50	2.82	1.64	2.87
2008	7.03	2.81	2.02	3.77
2009	7.23	3.29	2.95	6.47
2010	8.05	3.54	2.64	5.23
2011	7.85	4.07	3.13	5.96
2012	6.87	3.72	2.66	3.94
2013	6.88	4.15	2.72	3.95
2014	6.94	3.85	2.23	2.85
2015	9.01	3.71	2.76	2.92
2016	9.11	3.88	2.75	3.86
2017	8.18	4.15	2.82	5.20
2018	7.20	4.68	2.90	9.39
2019	6.59	3.63	2.84	7.01
2020	7.41	3.61	3.22	4.96
2021	8.01	4.12	3.83	5.44

Table 4a: Survey abundance estimates and associated standard errors in thousand tons for *M. paradoxus* for the depth range 0-500m for the South Coast and for the West Coast (Fairweather, 2022).

Year	West coast				South coast			
	Summer		Winter		Spring (Sept)		Autumn (Apr/May)	
	Biomass	(s.e.)	Biomass	(s.e.)	Biomass	(s.e.)	Biomass	(s.e.)
1985	168.989	(37.765)	290.281	(63.295)	-	-	-	-
1986	202.334	(37.745)	147.378	(21.667)	11.280	(3.111)	-	-
1987	284.434	(54.165)	180.158	(39.047)	16.381	(3.033)	-	-
1988	138.534	(20.303)	252.121	(71.246)	-	-	28.293	(8.673)
1989	-	-	434.092	(142.716)	-	-	-	-
1990	307.615	(87.841)	205.704	(43.607)	-	-	-	-
1991	331.177	(81.633)	-	-	-	-	27.570	(8.153)
1992	225.755	(33.711)	-	-	-	-	25.036	(6.650)
1993	340.079	(51.427)	-	-	-	-	162.375	(81.691)
1994	333.499	(56.259)	-	-	-	-	108.179	(38.369)
1995	317.104	(76.709)	-	-	-	-	70.890	(39.330)
1996	474.270	(92.744)	-	-	-	-	68.859	(19.929)
1997	543.615	(96.043)	-	-	-	-	121.707	(51.507)
1998	-	-	-	-	-	-	-	-
1999	542.830	(110.541)	-	-	-	-	263.256	(59.439)
2000	-	-	-	-	-	-	-	-
2001	-	-	-	-	16.668	(7.159)	-	-
2002	251.820	(32.690)	-	-	-	-	-	-
2003	386.321	(63.565)	-	-	98.434	(42.249)	185.345	(82.188)
2004	271.540	(55.710)	-	-	70.001	(22.156)	39.822	(22.153)
2005	296.065	(42.409)	-	-	-	-	26.691	(6.017)
2006	316.247	(57.332)	-	-	68.507	(18.283)	34.868	(5.843)
2007	407.377	(77.222)	-	-	66.267	(21.966)	102.195	(53.688)
2008	238.143	(37.018)	-	-	25.661	(8.324)	33.034	(9.340)
2009	310.760	(27.768)	-	-	-	-	45.030	(15.551)
2010	576.848	(88.202)	-	-	-	-	46.938	(12.160)
2011	380.185	(128.013)	-	-	-	-	21.054	(6.531)
2012	405.865	(59.099)	-	-	-	-	-	-
2013	136.260	(25.116)	-	-	-	-	-	-
2014	269.482	(37.492)	-	-	-	-	62.925	(24.802)
2015	207.583	(24.057)	-	-	-	-	111.411	(51.852)
2016	312.876	(33.250)	-	-	-	-	94.177	(51.731)
2017	319.024	(58.766)	-	-	-	-	-	-
2018	-	-	-	-	-	-	-	-
2019	243.560	(51.558)	-	-	-	-	-	-
2020	243.090	(43.989)	-	-	-	-	33.176	(15.444)
2021	-	-	-	-	-	-	-	-
2022	-	-	-	-	-	-	50.060	(20.430)

Table 4b: Survey abundance estimates and associated standard errors in thousand tons for *M. capensis* for the depth range 0-500m for the South Coast and for the West Coast (Fairweather, 2022).

Year	West coast				South coast			
	Summer		Winter		Spring (Sept)		Autumn (Apr/May)	
	Biomass	(s.e.)	Biomass	(s.e.)	Biomass	(s.e.)	Biomass	(s.e.)
1985	102.929	(18.888)	159.198	(18.982)	-	-	-	-
1986	113.154	(23.474)	115.218	(19.733)	96.768	(10.737)	-	-
1987	75.438	(9.709)	83.050	(10.306)	137.008	(13.057)	-	-
1988	66.365	(9.930)	48.046	(9.574)	-	-	154.548	(23.984)
1989	-	-	294.740	(67.495)	-	-	-	-
1990	400.142	(97.102)	156.337	(22.507)	-	-	-	-
1991	67.565	(9.656)	-	-	-	-	276.607	(25.274)
1992	95.401	(11.892)	-	-	-	-	124.495	(13.600)
1993	93.613	(14.390)	-	-	-	-	144.551	(12.379)
1994	124.497	(37.845)	-	-	-	-	153.790	(20.310)
1995	193.292	(24.270)	-	-	-	-	222.464	(31.245)
1996	87.969	(9.866)	-	-	-	-	222.176	(23.144)
1997	252.606	(42.721)	-	-	-	-	163.163	(17.274)
1998	-	-	-	-	-	-	-	-
1999	188.624	(31.362)	-	-	-	-	171.946	(13.330)
2000	-	-	-	-	-	-	-	-
2001	-	-	-	-	117.590	(20.093)	-	-
2002	105.093	(16.130)	-	-	-	-	-	-
2003	73.020	(12.518)	-	-	73.604	(9.142)	117.538	(17.192)
2004	194.294	(30.714)	-	-	96.933	(13.936)	92.796	(11.318)
2005	63.363	(11.498)	-	-	-	-	68.672	(5.302)
2006	73.655	(17.255)	-	-	92.831	(8.998)	116.298	(11.931)
2007	73.230	(9.306)	-	-	67.937	(6.553)	65.935	(5.303)
2008	52.577	(7.069)	-	-	87.836	(9.723)	102.169	(9.681)
2009	140.437	(26.486)	-	-	-	-	111.191	(10.832)
2010	162.402	(34.891)	-	-	-	-	170.261	(33.235)
2011	89.095	(23.574)	-	-	-	-	105.424	(10.688)
2012	84.746	(8.331)	-	-	-	-	-	-
2013	30.383	(4.575)	-	-	-	-	-	-
2014	219.756	(60.342)	-	-	-	-	63.389	(6.415)
2015	65.086	(9.178)	-	-	-	-	76.059	(6.873)
2016	115.058	(30.400)	-	-	-	-	83.197	(6.600)
2017	69.289	(14.486)	-	-	-	-	-	-
2018	-	-	-	-	-	-	-	-
2019	62.560	(7.697)	-	-	-	-	-	-
2020	109.983	(11.836)	-	-	-	-	132.099	(14.486)
2021	-	-	-	-	-	-	-	-
2022	-	-	-	-	-	-	89.119	(10.798)

Table 5: Key assessment outputs for the three OMs. Biomass units are thousand mt.

<i>M. paradoxus</i>	Run1	Run2	Run3	<i>M. capensis</i>	Run1	Run2	Run3
K^{sp}	467	471	459	K^{sp}	235	241	238
B_{MSY}	74	69	75	B_{MSY}	66	71	70
B_{MSY} / K^{sp}	0.16	0.15	0.16	B_{MSY} / K^{sp}	0.28	0.30	0.29
MSY	131	136	135	MSY	83	84	83
B_{2021}^{sp}	123	112	119	B_{2021}^{sp}	182	186	183
B_{2021}^{sp} / K^{sp}	0.26	0.24	0.26	B_{2021}^{sp} / K^{sp}	0.78	0.77	0.77
B_{2021}^{sp} / B_{MSY}	1.67	1.63	1.60	B_{2021}^{sp} / B_{MSY}	2.74	2.61	2.63
B_{2022}^{sp}	-	-	112	B_{2022}^{sp}	-	-	195
B_{2022}^{sp} / K^{sp}	-	-	0.24	B_{2022}^{sp} / K^{sp}	-	-	0.82
B_{2022}^{sp} / B_{MSY}	-	-	1.50	B_{2022}^{sp} / B_{MSY}	-	-	2.81

Table 6a: Negative log-likelihood values for three OMs. The values for Run3 in grey text indicate that those negative log-likelihood components are not comparable those from the other two runs on account of data updates.

Run	GLM CPUE	ICSEAF CPUE	Survey abun.	Comm. CAL	Survey CAL	Recruit. resid.	ALKs	Penalties	Total (w/o pen.)
Run1: OMP-2022 RC	-230.07	-39.26	-40.90	-1573.61	-1651.86	9.43	128.58	0.03	-3397.69
Run2: Longline update	-229.82	-39.34	-40.85	-1573.89	-1651.67	9.38	128.57	0.02	-3397.62
Run3: 2022 data update	-242.64	-39.08	-42.28	-1573.51	-1652.96	9.60	128.83	0.02	-3412.02

Table 6b: The first row of this Table lists the negative log-likelihood for Run1, while the remaining two rows show the difference between Run1 and the other two OMs. A positive value indicates a worse negative log-likelihood for the update, and these cells have also been shaded in grey. Grey text and dashed lines have been used to indicate the likelihood components for Run3 which are not comparable with those from Run1, on account of data updates.

Run	GLM CPUE	ICSEAF CPUE	Survey abun.	Comm. CAL	Survey CAL	Recruit. resid.	ALKs	Penalties	Total (w/o pen.)
Run1: OMP-2022 RC	-230.07	-39.26	-40.90	-1573.61	-1651.86	9.43	128.58	0.03	-3397.69
Run2: Longline update	0.25	-0.08	0.05	-0.27	0.19	-0.05	-0.01	-0.00	0.06
Run3: 2022 data update	-12.57	0.19	-1.39	0.11	-1.10	0.17	0.25	-0.00	-14.34

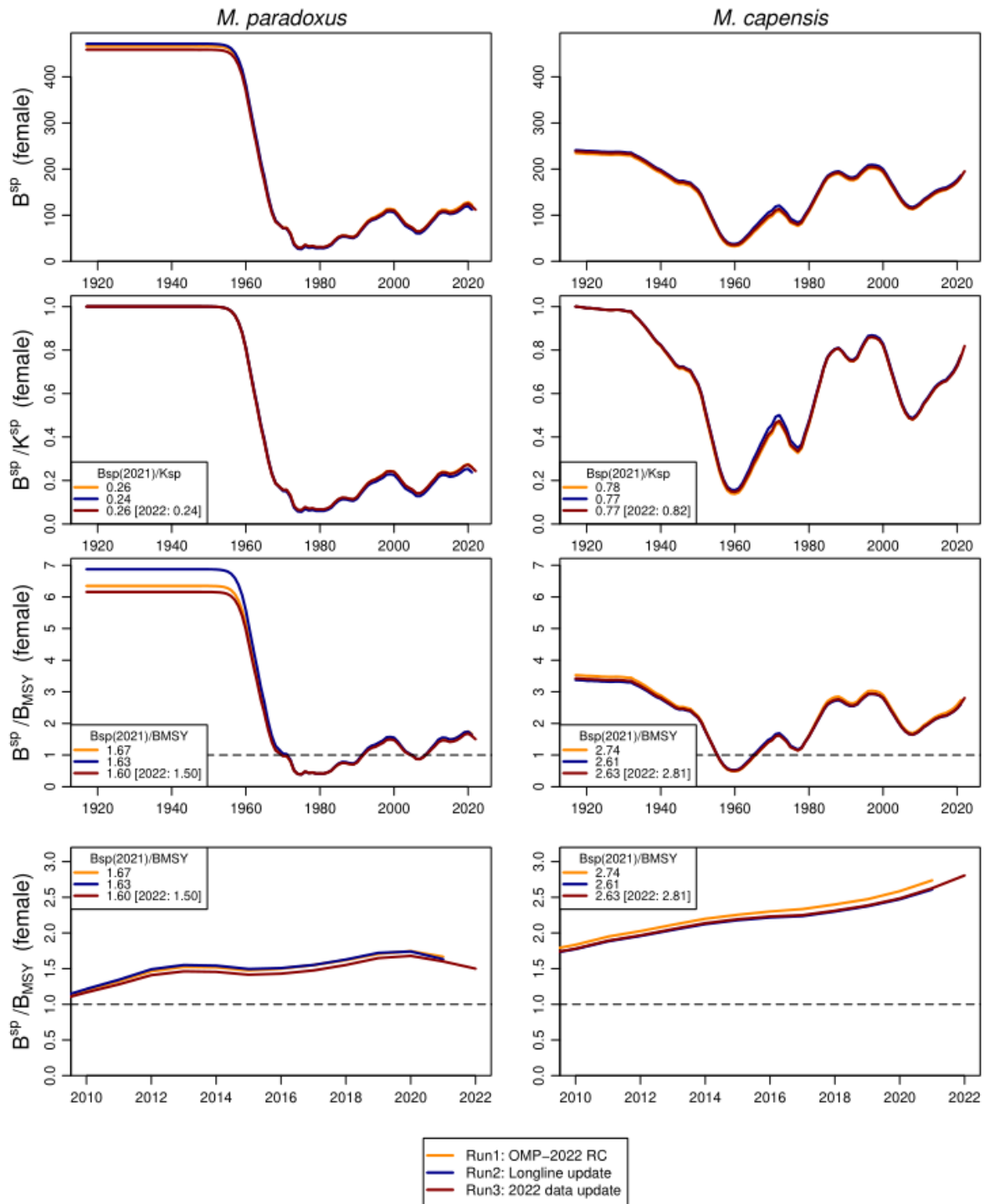


Figure 1: Female spawning biomass is shown for each species in absolute terms (top row), relative to pristine spawning biomass (second row), relative to B_{MSY} (third row) and relative to B_{MSY} but for the 2010-2022 time period only (fourth row) and. The depletion levels have been listed in the figure legends for the year 2021, with the 2022 values for Run3 included in square brackets.

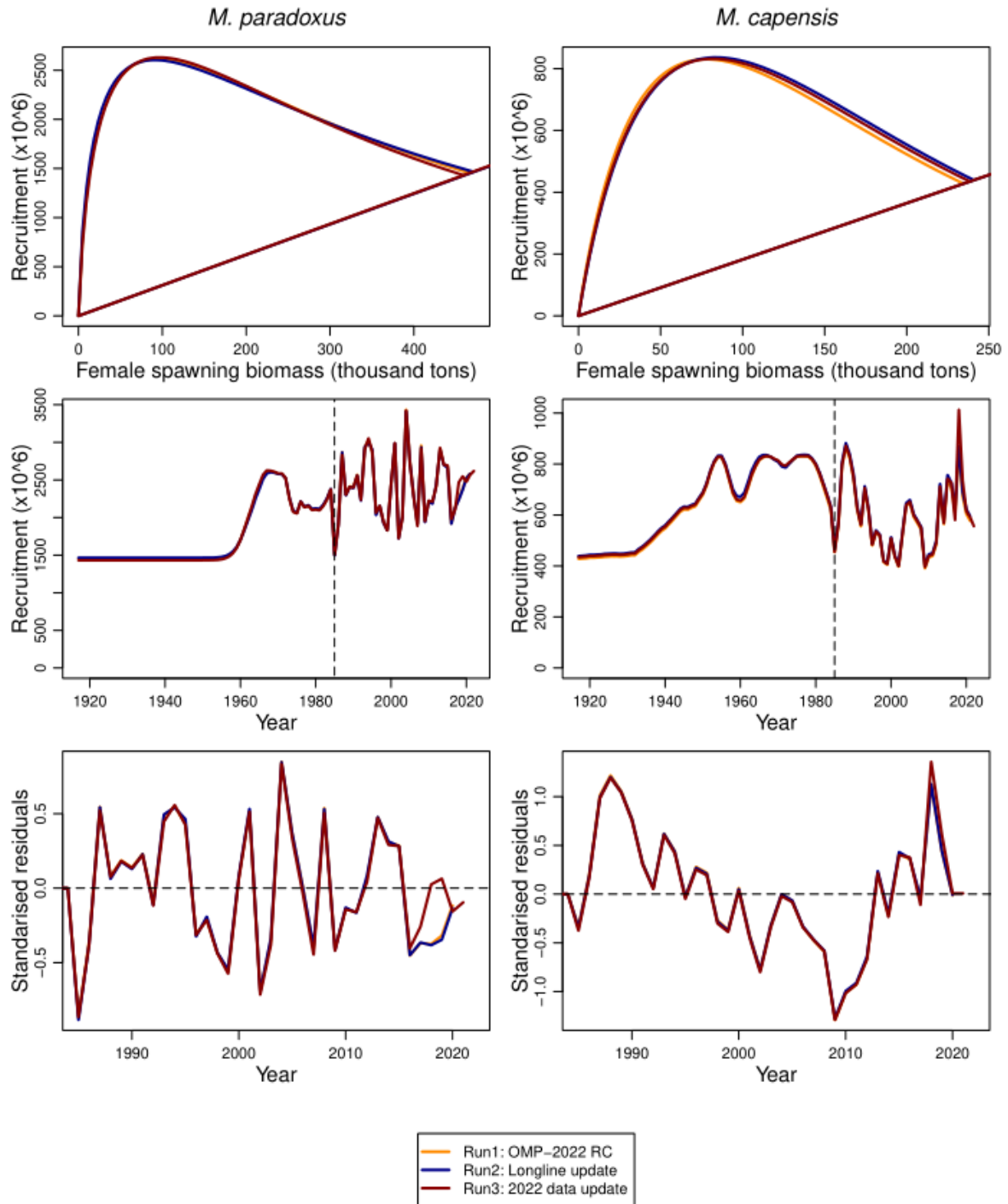


Figure 2: The top row shows the estimated recruitment-spawning biomass relationship for the three assessment models, while the second row shows the estimated recruitment values and the bottom row the standardized estimated recruitment residuals.

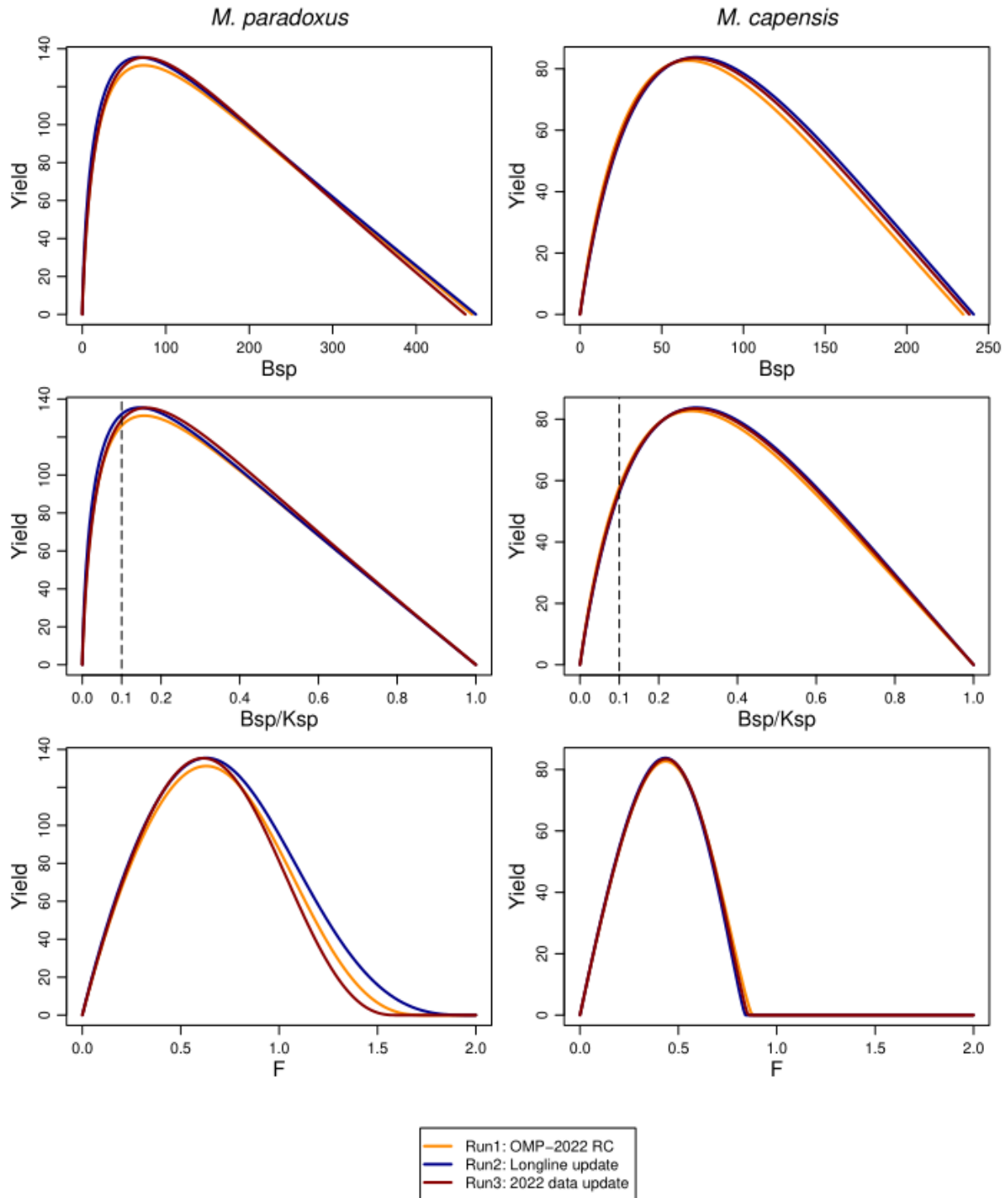


Figure 3: The estimated yield curve (yield in thousand tons) is plotted against female spawning biomass (top panel), female spawning biomass relative to pristine levels (middle panel) and fishing mortality rate F (bottom panel).

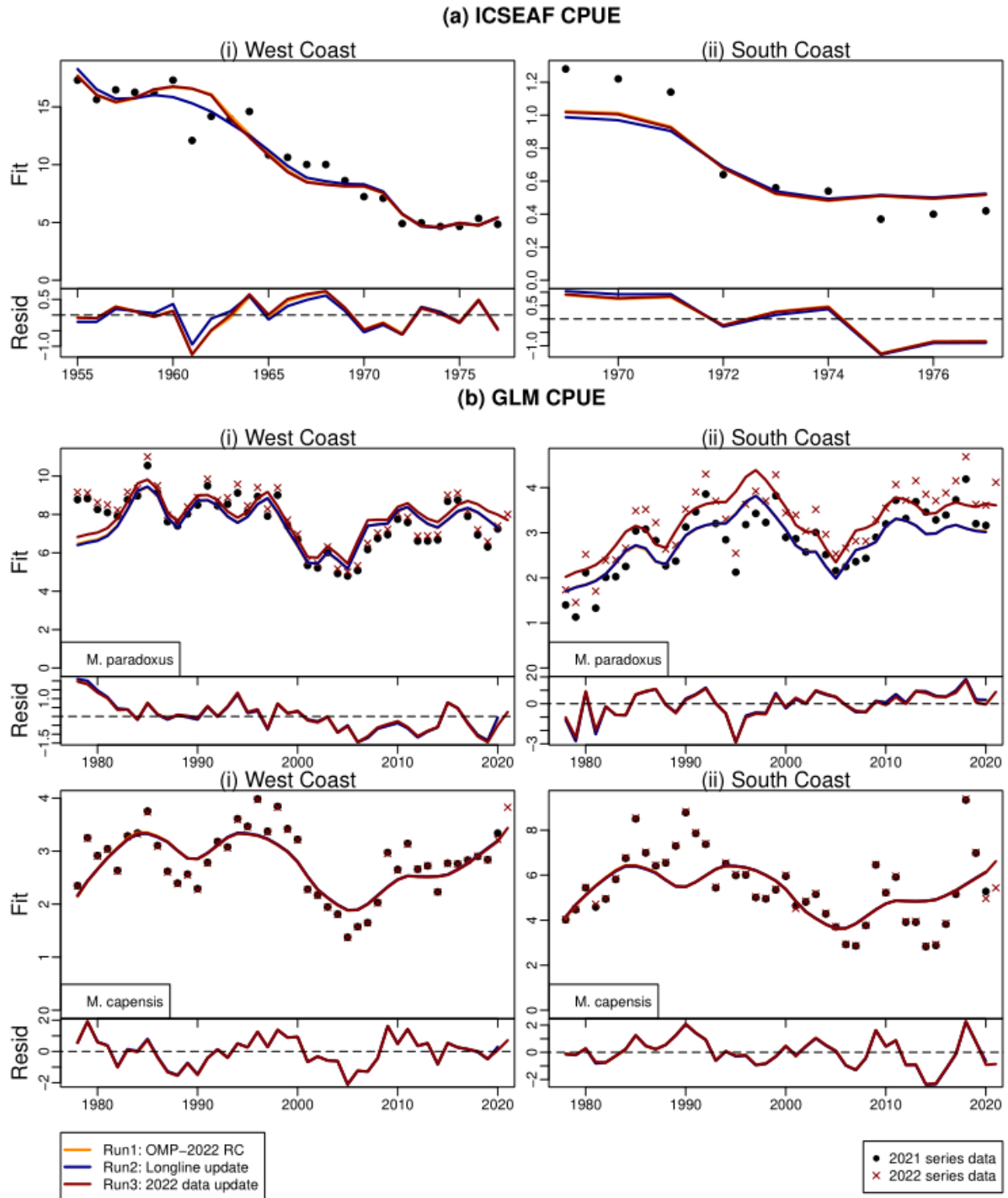


Figure 4: Fits to (A) the historical ICSEAF CPUE data and (B) the commercial GLM-standardized CPUE data are shown. Standardised residuals are shown underneath each plot. The updated 2022 GLM CPUE series has been shown by the red crosses.

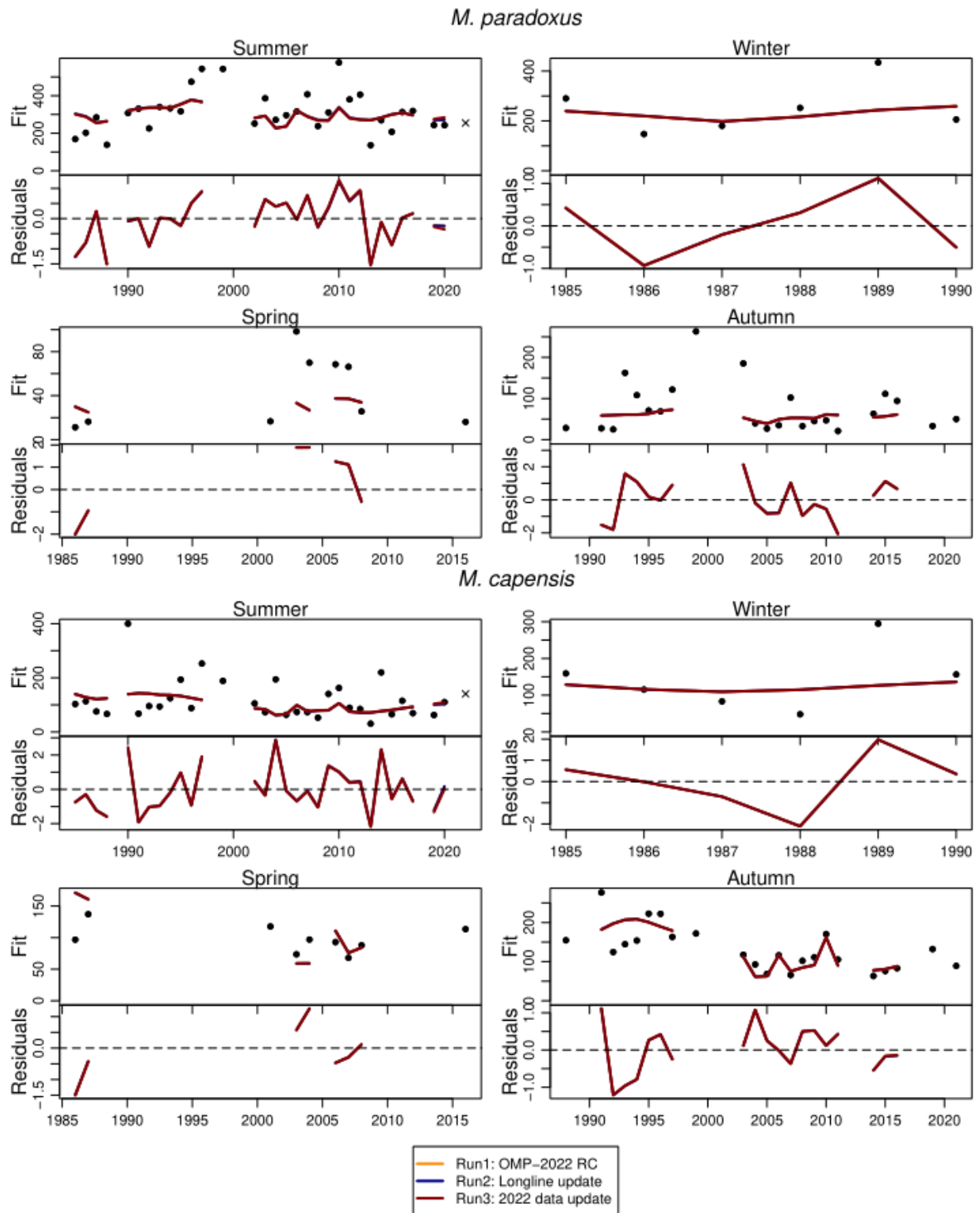


Figure 5: Fits to the survey relative abundance series are shown for the three OMs. Standardised residuals are shown underneath each plot. The new 2022 summer survey data points are shown by the crosses. Note that at this level of resolution, the three fits can scarcely be distinguished.

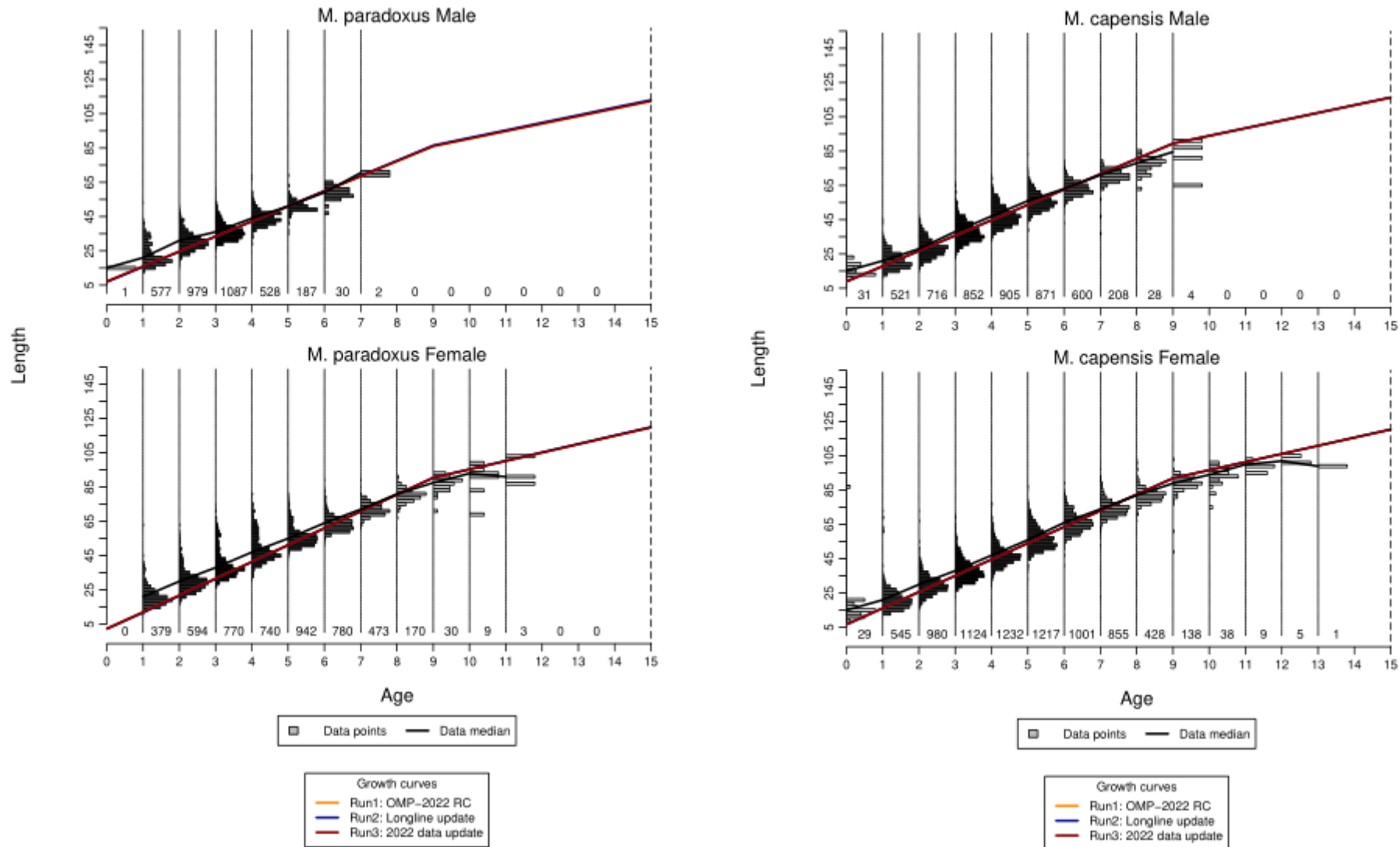


Figure 6: The estimated growth curves are superimposed on the length-at-age distributions derived from the age-length-key data. The numbers printed at the bottom of each age class indicated the number of data points available, and the shading of the bars is relative to these numbers, with darker shades indicating more data points.